Data Science Project
(Spam Prediction on Spare Text Dataset)

By
Jabez Solomon J

Introduction

- Spam emails are a major issue, consuming resources and posing security risks.
- Spam can lead to financial losses and identity theft.
- Legitimate emails can be marked as spam, impacting marketing efforts.
- Enforcing anti-spam laws is difficult due to international spammers.
- Machine learning helps identify and filter spam emails.
- Gmail uses advanced machine learning algorithms and user feedback to combat spam effectively.
- Benefits of Machine Learning are Improved efficiency, accuracy, and security in email communication.

Python Library

The below mentioned python libraries are the implemented library for this project:

1. Pandas
2. Matplotlib
3. Seaborn
4. Joblib
5. Openpyxl
6. Re
7. Sklearn
8. Streamlit

Data Preprocessing

- In our email analysis project, the dataset comprises a substantial volume of data with 5173 rows and 3002 columns.

- An overview of the key preprocessing steps and their significance:

1. Exclusion of Non-Feature Columns:

- Removed 'Email No.' and 'Prediction' columns as they are not part of the feature set.

2. Email Length Calculation:

- Computed 'email_length' for each email to measure its length based on word frequency or attributes.

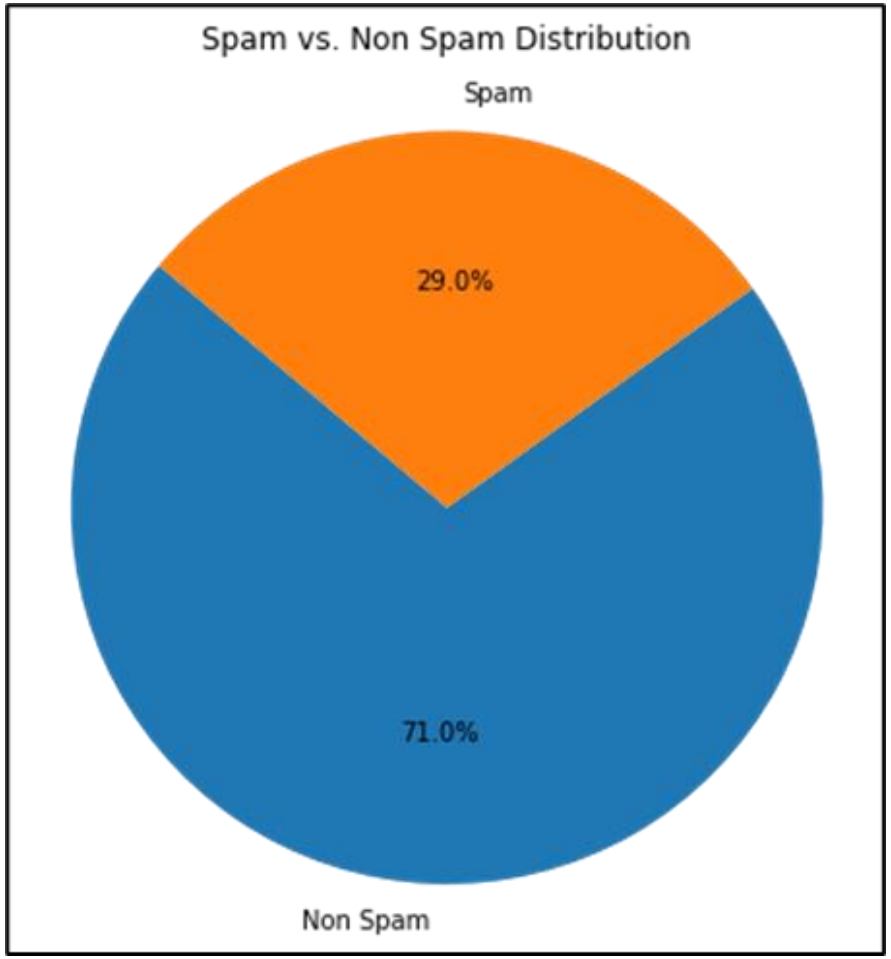
3. Count of Capitalized Words:

- Calculated 'capitalization_count' to identify the number of words starting with capital letters.

4. Word Count Calculation:

- Computed 'word_count' by counting unique words in each email, treating non-zero entries as individual words.

Data Preprocessing(Cont..)



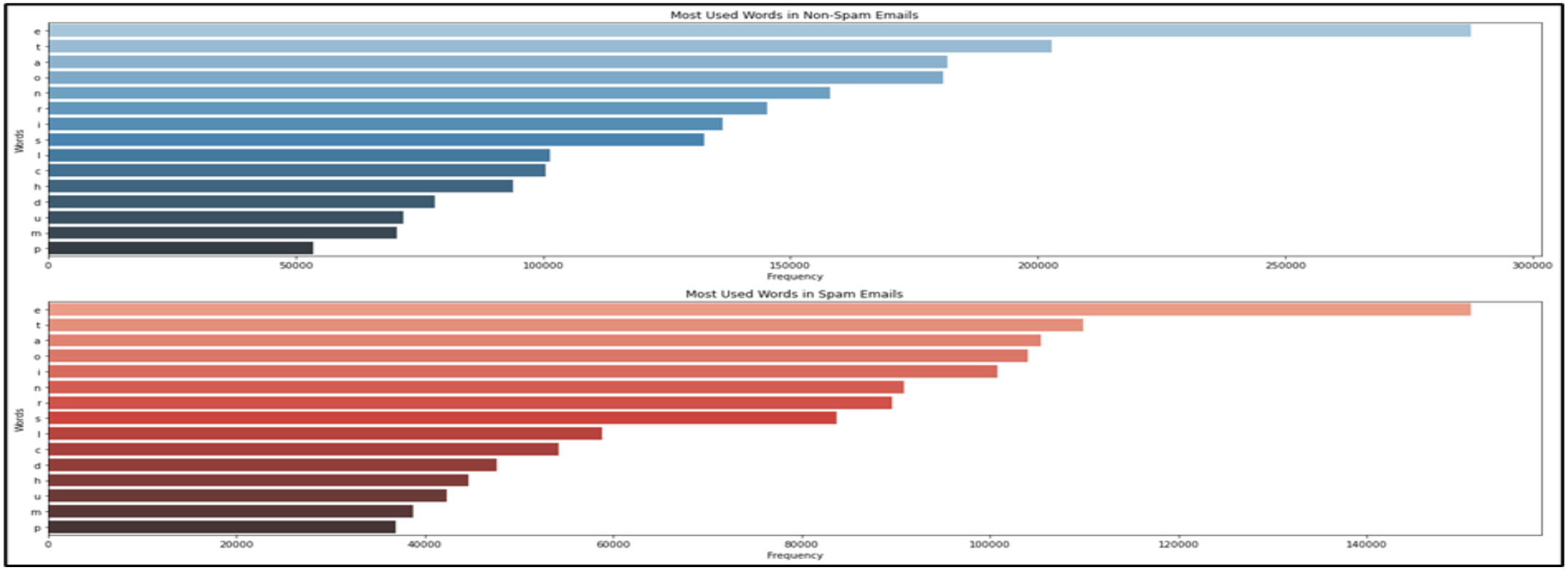
1. **Class Distribution Analysis:**
- 'Prediction' column values: '0' for Non-Spam, '1' for Spam.
- Pie chart shows a significant class distribution imbalance: 71% Non-Spam, 29% Spam.

2. Impact of Imbalanced Data:
- Imbalanced data can bias machine learning models towards the majority class (Non-Spam).
- This bias can result in less accurate detection of Spam emails.

3. Addressing Imbalance:
- Techniques like resampling, anomaly detection, and class weighting can mitigate the issue.

4. Importance of Representative Data:
- Highlighting the need to train machine learning models on representative datasets.
- The pie chart serves as a visual reminder to improve preprocessing techniques for better model performance despite imbalanced data challenges.

Data Preprocessing(Cont..)



Data Preprocessing(Cont..)

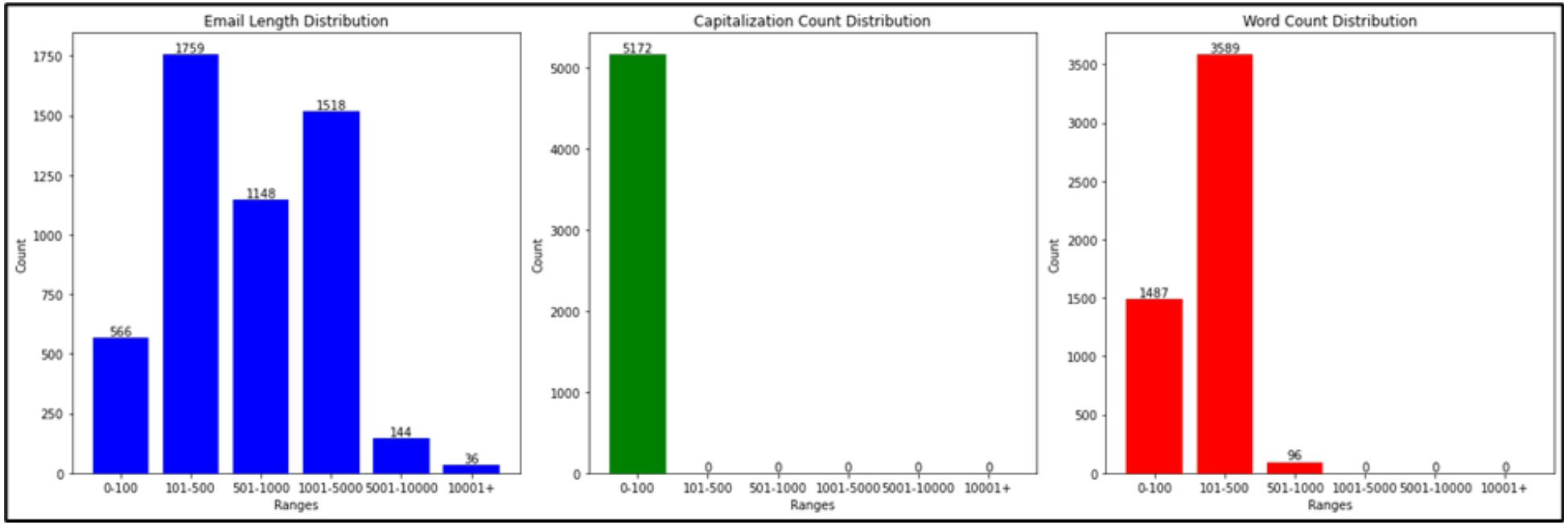
1. **Word Analysis in Emails:**

- Examined the most frequently used words in both spam and non-spam emails.
- Visualizations showcase the top fifteen words in each category.

2. **Distinguishing Features:**

- "Most Used Words in Non-Spam Emails" bar graph (shades of blue) highlights words typical in professional or formal communication.
- "Most Used Words in Spam Emails" graph (shades of red) reveals distinct terms associated with urgency, action, or sales, common in unsolicited promotional content.

Data Preprocessing(Cont..)



Data Preprocessing(Cont..)

1. Graph Generation with Seaborn:

- Visualizations were created using Seaborn's barplot function.
- These visualizations offer a clear contrast between two email types: spam and non-spam.
- The x-axis represents word frequency, while the y-axis lists the words themselves.

2. Data Distribution Visualizations:

- To understand the distribution of "email_length," "capitalization_count," and "word_count," data was categorized into bins and visually presented.

3. Email Length Distribution (Blue):

- Majority of emails fall within the 101-500 and 501-1000 character ranges.
- A significant drop is observed as character count increases, indicating concise emails.

Data Preprocessing(Cont..)

4. Capitalization Count Distribution (Green):

- Most emails exhibit capitalization counts between 0-100, reflecting standard capitalization usage.
- Few emails contain excessive capitalized words, potentially indicative of promotional content.

5. Word Count Distribution (Red):

- Predominance of emails in the 101-500 word range, followed by the 0-100 word range.
- A decline in frequency as word count increases, suggesting shorter emails in the dataset.

6. Data Binning and Visualization:

- Data was segmented into meaningful ranges using custom-defined bins.
- Matplotlib was used to create bar graphs for visualizing data distribution within these bins.
- Graphs include count annotations for easy interpretation.

Machine Learning

1. Model Selection for Classification:

- Two machine learning models were chosen for comparative analysis: Naive Bayes and Random Forest.

2. Naive Bayes Classifier:

- Well-suited for text classification due to simplicity, efficiency, and probabilistic foundation.
- Strong choice for high-dimensional text data.
- Handles both binary and multi-class classification effectively.
- Assumes feature independence and is robust to irrelevant features.

3. Random Forest Classifier:

- An ensemble learning method known for high accuracy and robustness.
- Aggregates multiple decision trees to reduce overfitting.
- Effective for datasets with imbalanced classes, such as spam and non-spam emails.
- Ranks feature importance, providing valuable insights into spam detection patterns.

Naïve Bayes Classifier

1. Naive Bayes Model Evaluation:

- Thorough evaluation of the Naive Bayes-based email classification model.

2. High Accuracy:

- Model achieved an accuracy of approximately 93.56%, demonstrating precise classification.

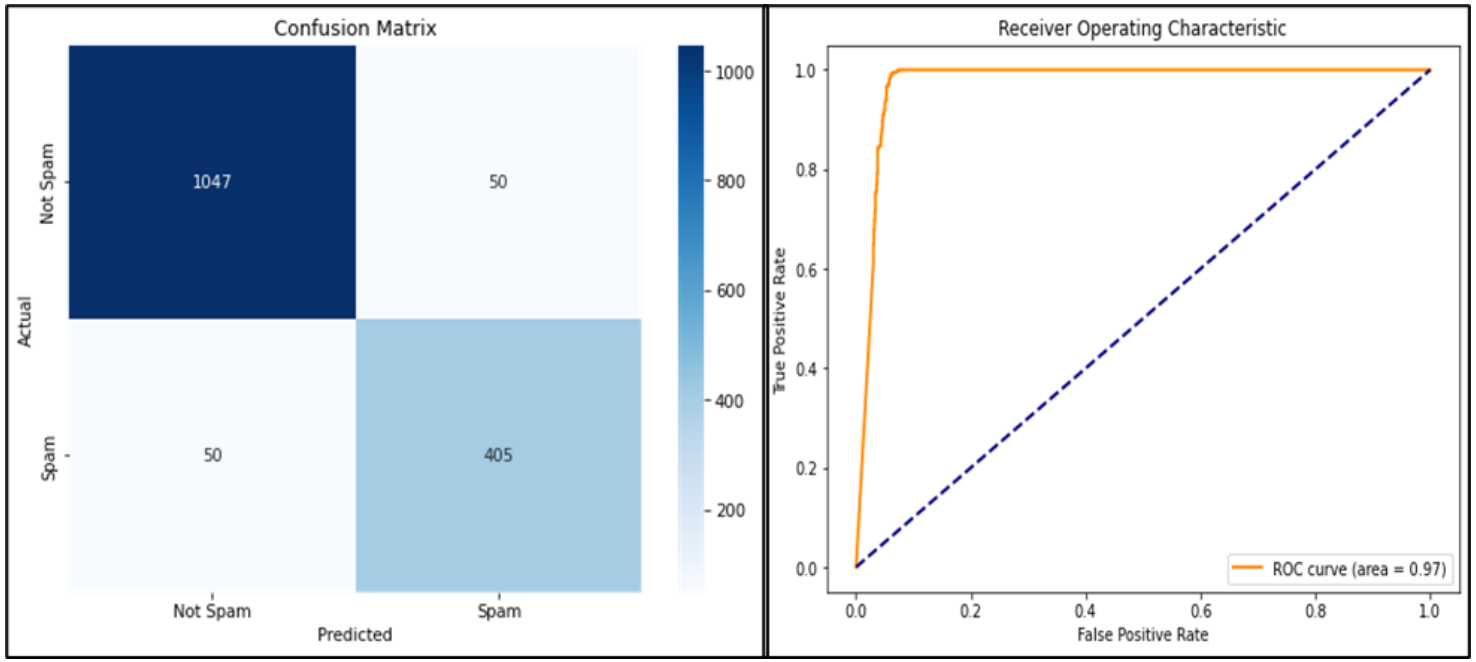
```
Model Accuracy: 0.9355670103092784

Performance Report:
              precision    recall  f1-score   support

           0       0.95      0.95      0.95      1097
           1       0.89      0.89      0.89       455

    accuracy                           0.94      1552
   macro avg       0.92      0.92      0.92      1552
weighted avg       0.94      0.94      0.94      1552
```

Naïve Bayes Classifier(Cont..)



3. Precision and Recall for Non-Spam:

- Precision and recall both at 0.95 for Non-Spam (label '0'), indicating 95% correctness in predictions and 95% identification of actual non-spam emails.

4. Precision and Recall for Spam:

- Precision and recall at 0.89 for Spam (label '1'), showcasing effective identification of spam emails.

Naïve Bayes Classifier(Cont..)

5. Excellent F1-Scores:

- F1-score at 0.95 for non-spam and 0.89 for spam, highlighting excellent model performance.

6. High AUC in ROC Curve:

- ROC curve has an AUC of 0.97, indicating exceptional ability to differentiate between spam and non-spam with high true positive rate and low false positive rate.

7. Balanced Confusion Matrix:

- 1,047 non-spam emails correctly classified with only 50 false positives.
- 405 spam emails correctly identified with the same number of false negatives.
- Balanced performance despite imbalanced data challenges.

8. Viable Spam Classification Tool:

- The Naive Bayes classifier demonstrates high accuracy and balanced precision-recall, making it a valuable tool for spam email classification.

Random Forest Classifier

Accuracy Score: 0.978743961352657

Performance Report:

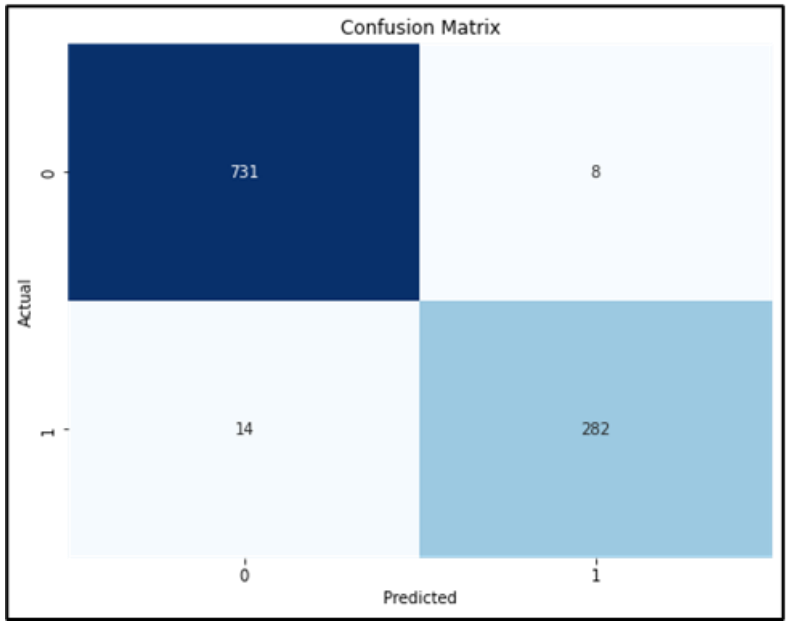
	precision	recall	f1-score	support
0	0.98	0.99	0.99	739
1	0.97	0.95	0.96	296
accuracy			0.98	1035
macro avg	0.98	0.97	0.97	1035
weighted avg	0.98	0.98	0.98	1035

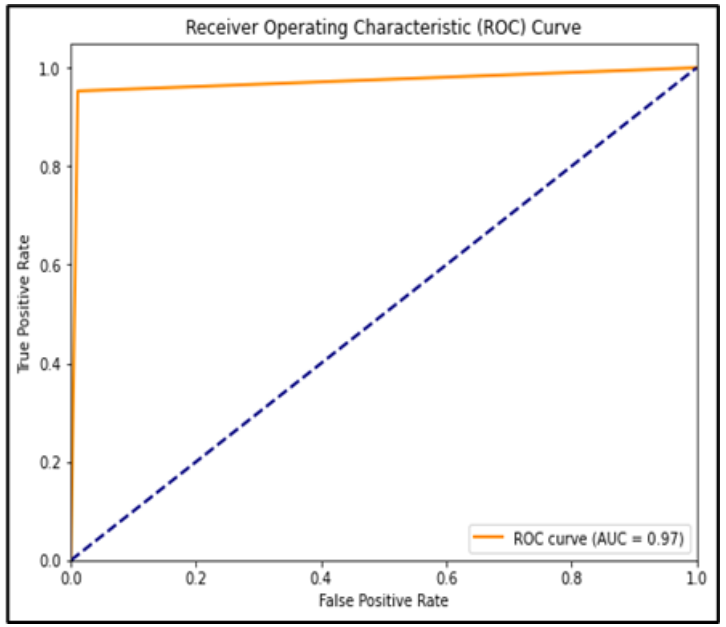
1. Random Forest Model Evaluation:

- Applied the Random Forest algorithm to the preprocessed email dataset and conducted a performance evaluation.

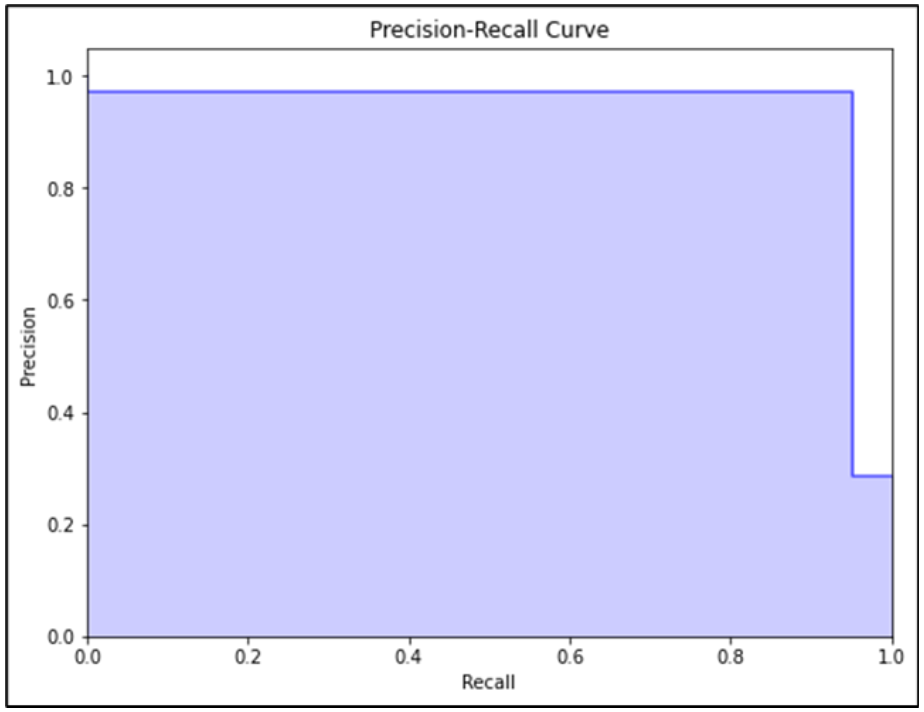
2. High Accuracy:

- Achieved an accuracy score of approximately 97.87%, a significant improvement over Naive Bayes.





Random Forest Classifier(Cont..)



3. Precision, Recall, and F1-scores:

- Non-Spam (label '0') precision of 0.98, recall of 0.99, and F1-score of 0.99, indicating high precision and reliability.
- Spam (label '1') precision of 0.97, recall of 0.95, and F1-score of 0.96, showcasing accurate identification of spam emails.

4. Precision-Recall Curve:

- Demonstrates high precision across all recall levels for spam detection, ideal for minimizing false positives in spam classification.

5. ROC Curve and AUC:

- ROC curve with an AUC of 0.97, confirming excellent classification capabilities, identical to Naive Bayes.

6. Balanced Confusion Matrix:

- Correctly classified 731 non-spam emails with only 8 false positives.
- Identified 282 spam emails with 14 false negatives, highlighting balanced accuracy in classifying both classes.

Conclusion of First Attempt

1. Random Forest Outperforms Naive Bayes:

- Random Forest exhibits superior performance metrics compared to Naive Bayes, including accuracy, precision, recall, and F1-score.

- Both models have high AUC values, indicating excellent classification abilities.

2. Edge of Random Forest:

- Random Forest excels due to higher precision and recall, especially in spam email classification.

3. Advantages of Random Forest:

- Random Forest handles dataset complexity and imbalances effectively.

- Offers interpretability through feature importance scores.

4. Preferred Model:

- Random Forest is the preferred model for email classification due to its higher accuracy and greater reliability in distinguishing between spam and non-spam emails.

Machine Learning Performance Boost

1. Strategic Dataset Refinement:
- Employed a data refinement strategy based on the distribution of "email_length" from the Distribution Graph.

2. Focus on Common Length Ranges:
- Focused on the '101-500', '501-1000', and '1001-5000' email length bins due to their higher frequency in the dataset, suggesting potential predictive features.

3. Selective Filtering:
- Filtered the main dataset to include records falling within the chosen length bins.
- The aim was to concentrate on characteristic email lengths, excluding outliers that might affect the model's learning process.

4. Refined Dataset Preservation:
- The refined dataset was exported to an Excel file named 'filtered_data.xlsx' for reproducibility and further analysis.
- Ensured traceability of steps in the data preprocessing stage.

5. Performance Re-evaluation:
- Re-evaluated machine learning models after concentrating on more typical email lengths.
- The objective was to assess any performance improvements, potentially driven by predictive features in common email length ranges.

Updated Naïve Bayes

1. Refined Naive Bayes Model:

- Achieved a strong accuracy score of approximately 95.33% after dataset refinement, signifying high precision in classification.

2. Precision, Recall, and F1-scores:

- For Non-Spam (label '0'), achieved a precision of 1.00 (no false positives), recall of 0.94, and F1-score of 0.97.

- For Spam (label '1'), achieved a precision of 0.86, recall of 1.00 (no false negatives), and an F1-score of 0.93.

3. Impressive ROC Curve:

- Maintained an impressive AUC of 0.98, affirming the model's exceptional ability to distinguish between spam and non-spam emails.

4. Updated Confusion Matrix:

- High numbers of true positives and true negatives, with 879 correctly classified non-spam emails and 387 correctly identified spam emails.

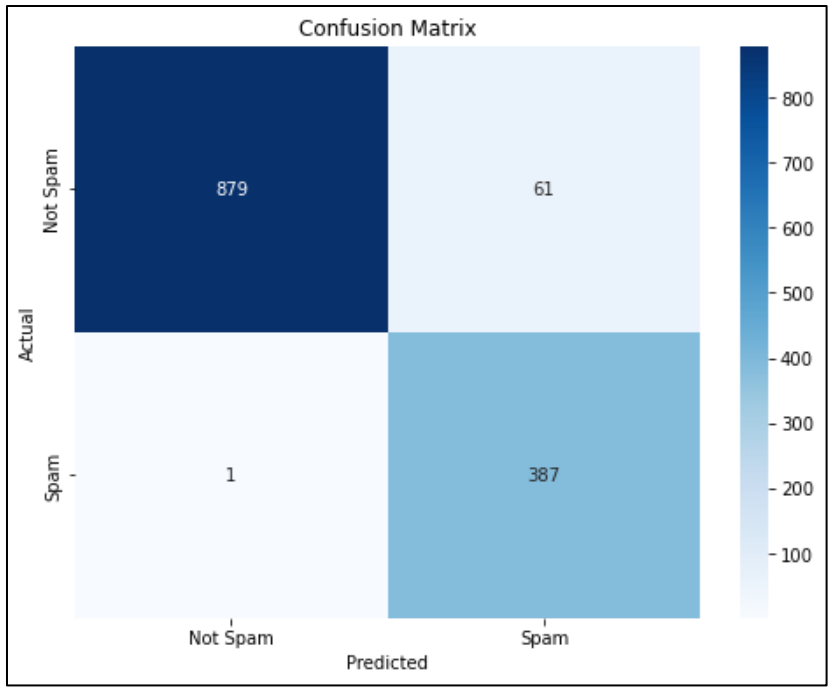
- Minimal false positives and false negatives, indicating the refined model's strong discernment abilities.

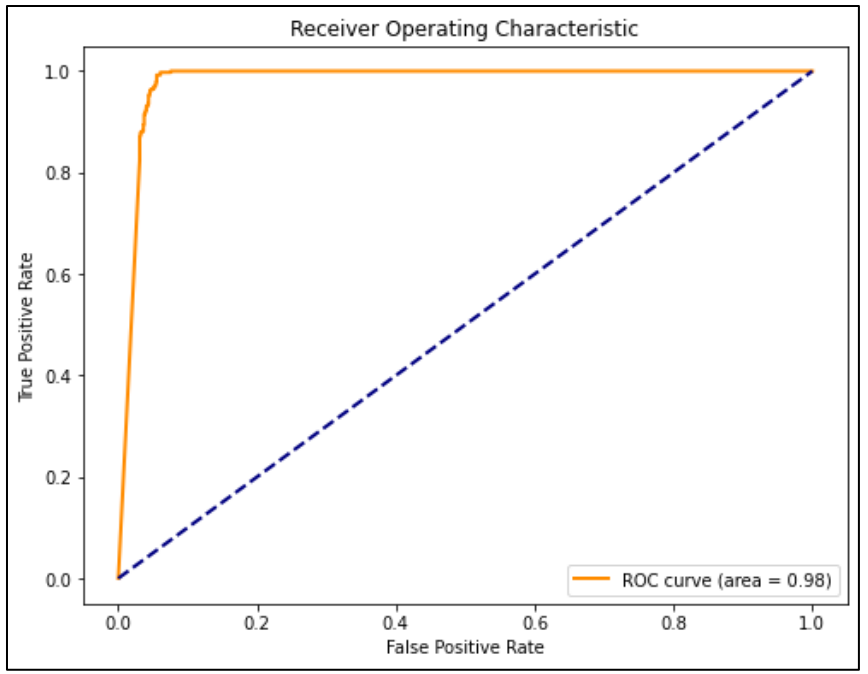
Updated Naïve Bayes(Cont..)

Model Accuracy: 0.9533132530120482

Performance Report:

	precision	recall	f1-score	support
0	1.00	0.94	0.97	940
1	0.86	1.00	0.93	388
accuracy			0.95	1328
macro avg	0.93	0.97	0.95	1328
weighted avg	0.96	0.95	0.95	1328





Updated Random Forest

1. Random Forest Model Performance:

- Achieved an accuracy score of approximately 96.94%, demonstrating proficient classification capabilities within the refined dataset.

2. Balanced Precision and Recall:

- Maintained a commendable balance between precision and recall, crucial for effective spam detection, where misclassification costs can be significant.

3. ROC Curve and AUC:

- ROC curve presented an AUC of 0.96, indicating strong discriminative ability between classes.

4. Updated Confusion Matrix:

- Accurately classified 605 Non-Spam emails and correctly identified 253 Spam emails.

- Minimal false positives (8) and false negatives (19), highlighting the model's reliability.

Updated Random Forest(Cont..)

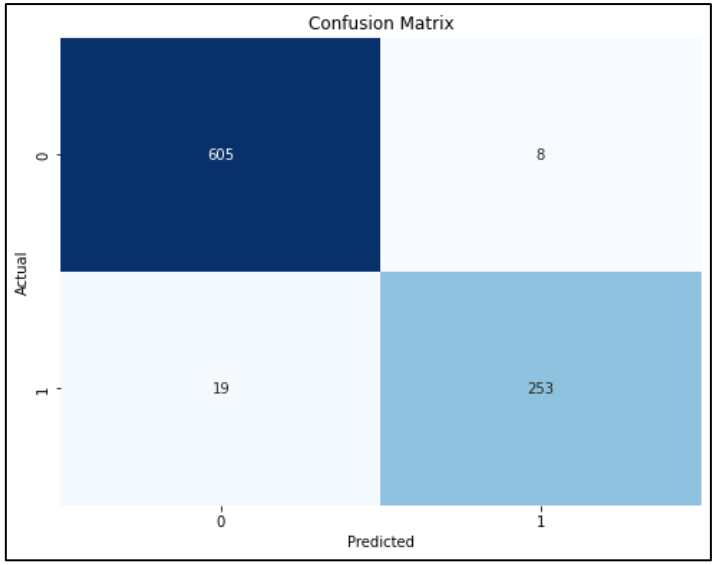
Accuracy Score: 0.9694915254237289

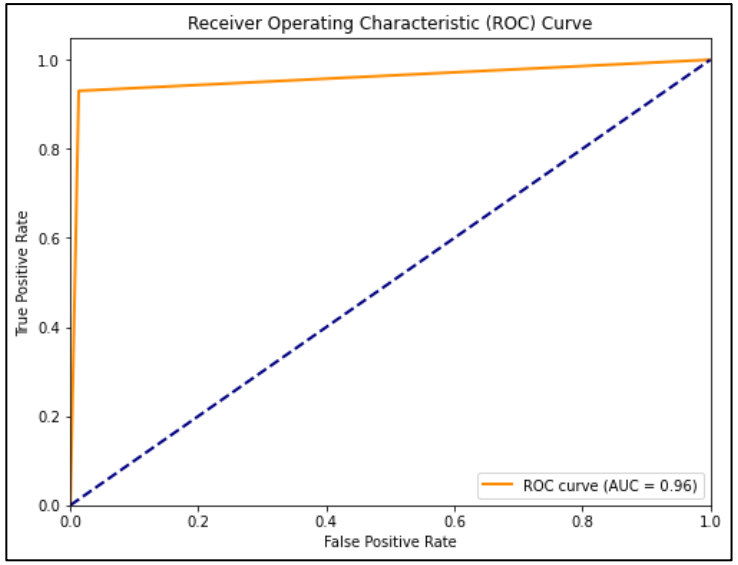
Performance Report:

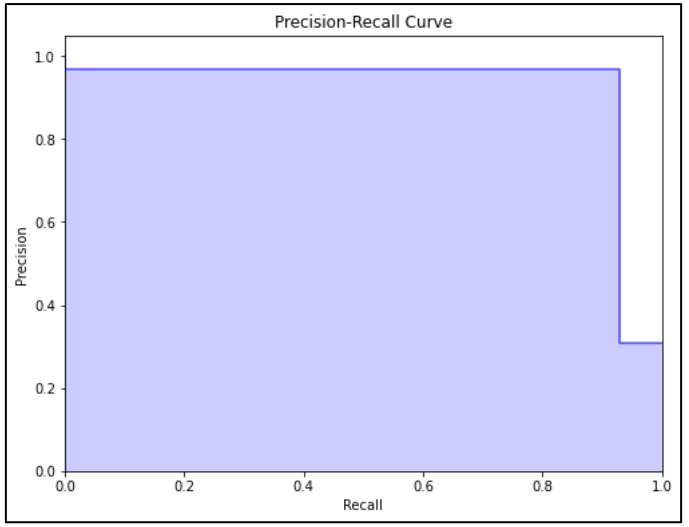
	precision	recall	f1-score	support
0	0.97	0.99	0.98	613
1	0.97	0.93	0.95	272
accuracy			0.97	885
macro avg	0.97	0.96	0.96	885
weighted avg	0.97	0.97	0.97	885

5. Performance Comparison:

- The refined Random Forest classifier showed a slight reduction in performance compared to previous results.
- Minor decreases in recall for Spam emails and AUC were observed.
- High accuracy and F1-scores suggest that the model remains robust and effective.







Updated Conclusion

1. Dataset Refinement Impact:

- Refining the dataset based on email length ranges did not significantly enhance the Random Forest classifier's performance.

2. Comparable Performance:

- Both Naive Bayes and Random Forest classifiers performed well, each with its respective strengths.

3. Random Forest Strength:

- The Random Forest classifier remains strong for email classification despite marginal metric decreases.

4. Model Selection:

- Random Forest was chosen as the preferred model through the entire machine learning process.

5. Model Availability:

- The trained Random Forest model has been saved as '**spam_classification_model.joblib**' for future use.

Spam Classification Web Application Development

1. Web Application for Spam Classification:
- Developed a user-friendly web application using Streamlit to operationalize the spam classification model.

2. Functionality Overview:
- Sidebar for file upload, accepting ".txt" files with email content.
- Text data cleaning and extraction of words.
- Comparison with a pre-existing Excel template representing the feature set.
- Button to trigger processing, model loading, and predictions.
- Results updated in a new Excel file for download.
- Display of predictions distinguishing spam and non-spam emails.

3. **Technical Implementation:**
- Functions for reading and preprocessing text, aligning words with features, and model loading and prediction.
- Seamless and interactive user experience, from file upload to download.

4. **Deployment and Usage:**
- Designed for ease of use, requiring only text file upload for instant spam classification.
- Display of results within the application, with an updated Excel file for download.
- Clear classification summary provided in the main content area (1 for spam and 0 for not spam).

Spam Classification Web Application Development(Cont..)

5. Practical Application:

- The web application represents a practical step towards applying the spam classification model in real-world scenarios.
- Offers accessible and immediate spam detection without requiring technical expertise.

